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Week 6 Baseline Report: ED Triage Classification

1. Introduction

This report documents the first machine-learning baselines built to predict Emergency Severity Index (ESI) triage level from information available at the point of triage. The purpose of a baseline is not sophistication but defensibility: a simple, transparent model that can be measured, explained to a clinician, and used as the floor against which any future, more complex model must prove its worth.

The dataset is the Yale EMMLC triage extract used throughout Weeks 5–6: 55,121 de-identified adult ED encounters, with vitals and chief-complaint flags as predictors and the nurse-assigned ESI level (1–5, most to least urgent) as the target. Outcome fields such as disposition and previousdispo are excluded from the feature set, as they are known only after triage and would leak future information into a triage-time prediction.

2. Methods

- Data split: 80/20 stratified train/test split on ESI, random_state = 42, to preserve class proportions across both sets and keep the split reproducible.
- Dummy Classifier: DummyClassifier (strategy = "stratified", random_state = 42) — guesses ESI levels at random, in the same proportions they occur in the data. This is the floor every real model must beat.
- Logistic Regression: features imputed (median) and scaled (StandardScaler) before training, since logistic regression is sensitive to feature magnitude. LogisticRegression(max_iter = 1000, random_state = 42).
- Decision Tree: DecisionTreeClassifier(max_depth = 5, random_state = 42), trained on unscaled features. A depth of 5 was chosen to keep the tree shallow enough to remain readable and explainable to a clinician, while still allowing enough splits to separate the five ESI classes.

3. Results

Model comparison

Model	Accuracy	Macro F1	Weighted F1	Recall (ESI 1)
Dummy Baseline	0.389	0.200	0.387	**0.000**
Logistic Regression	0.657	0.504	0.653	**0.333**
Decision Tree	0.571	0.222	0.480	**0.000**

Confusion matrices

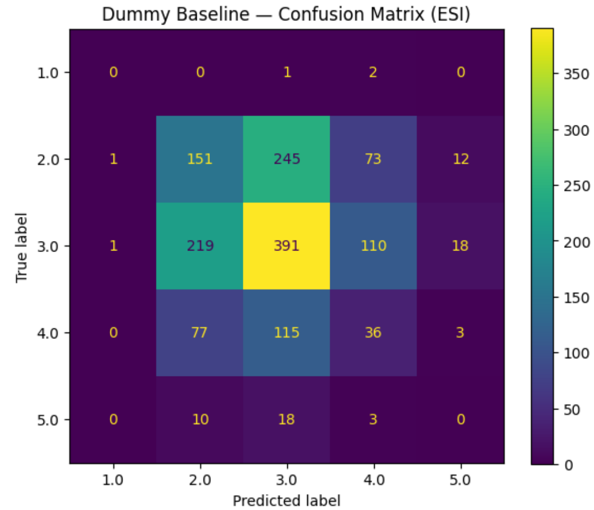

Figure 1: Dummy baseline — confusion matrix.

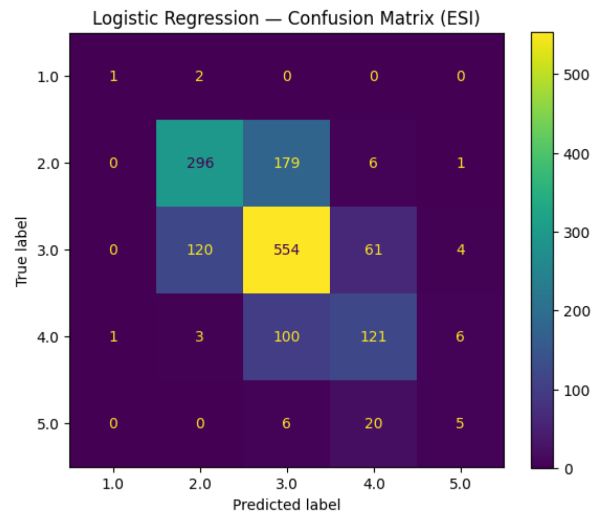

Figure 2: Logistic regression — confusion matrix.

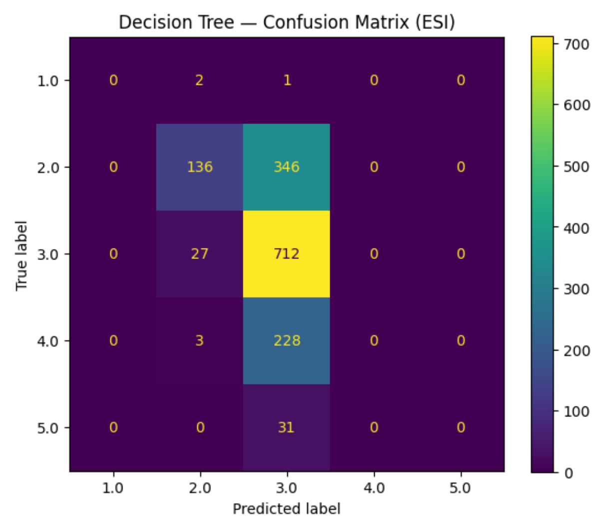

Figure 3: Decision tree — confusion matrix.

Macro vs. weighted F1

Macro F1 gives equal weight to every ESI level, regardless of how many patients fall into it — so performance on the rare ESI 1 class counts just as much as performance on the common ESI 3 class. Weighted F1 instead weights each class by how many patients it actually contains, so performance on frequent classes (ESI 2 and 3) dominates the score. The two diverge sharply for both real models: logistic regression scores 0.504 macro F1 but 0.653 weighted F1, and the decision tree scores 0.222 macro F1 but 0.480 weighted F1. This gap reveals that both models perform far better on common, mid-acuity cases than on rare, high-acuity ones (a distinction that overall accuracy), or weighted F1 alone, would hide entirely.

4. Primary Metric Justification

The primary metric for this baseline is recall for ESI Level 1. In emergency triage, missing a critically ill patient (under-triage) is far more dangerous than incorrectly flagging a stable patient as urgent (over-triage) — a false alarm costs a few extra minutes of clinician review, while a missed ESI 1 patient risks delayed resuscitation and potential death. Because ESI 1 is also the rarest class in this dataset, overall accuracy is a poor guide: a model can score highly by predicting the majority classes well while ignoring the sickest patients entirely. Recall for ESI 1 directly measures the one failure mode that matters most clinically — of all the truly critical patients, how many did the model actually catch.

5. Failure Mode Reflection

The most concerning failure mode in both baseline models is under-triage of ESI Level 1 patients. The decision tree recalls none of the ESI 1 cases in the test set, it misses every single critically ill patient, despite reasonable overall accuracy (0.571). Logistic regression performs only marginally better, catching about a third of ESI 1 cases (recall 0.333). A patient missed by either model would be routed to a lower-acuity queue, delaying the immediate resuscitation-level attention they need - a genuinely dangerous outcome in practice.

This failure is driven by severe class imbalance (only a handful of ESI 1 cases to learn from) rather than a flaw specific to either algorithm, and it is the clearest evidence that accuracy is the wrong metric to optimize for this task going forward.

6. Conclusion

Of the two real models, logistic regression is the stronger baseline: it achieves higher accuracy (0.657 vs 0.571), higher macro and weighted F1, and most importantly for this clinical task, a non-zero recall on ESI Level 1 (0.333 vs 0.000). The decision tree, despite being easier to visualize and explain, fails outright on the rare, most critical class.

Neither model is yet fit for clinical deployment: a one-in-three catch rate on the sickest patients is not acceptable for a system intended to support triage decisions. However, logistic regression is a reasonable starting point for further development. Future work should prioritize techniques that directly address class imbalance, such as class weighting, resampling, or a cost-sensitive threshold tuned specifically to raise ESI 1 recall, rather than pursuing model complexity for its own sake.